



Multi-phase fluid–structure interaction using adaptive mesh refinement and immersed boundary method

Pedro Ricardo C. Souza¹ · Hélio Ribeiro Neto¹ · Millena Martins Villar¹ · João Marcelo Vedovotto¹ · Aldemir Ap Cavalini Jr¹ · Aristeu Silveira Neto¹

Received: 19 August 2021 / Accepted: 4 February 2022 / Published online: 22 March 2022
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Abstract

In the present work, the authors aim to solve incompressible fluid–structure interaction problems in multi-phase flows using the partitioned approach with an in-house three-dimensional, block-structured, adaptive mesh refinement (AMR) code called MFSim. A projection method for large eddy simulation (LES) of Navier–Stokes equations and a VOF-PLIC (volume of fluid - piecewise linear interface calculation) method with finite volume discretization is used to simulate the turbulent incompressible multi-phase flows. The solid structures, immersed in the flow, have the fluid-dynamic forces modeled with the immersed boundary method and displacements calculated using the finite element model based on the Mindlin–Reissner plate theory. In the partitioned algorithm, fluid dynamic forces acting on the immersed surface are applied to the finite element model to predict the structure’s displacements and velocities at each time step. Strong coupling between fluid and structure was implemented and the advantages and drawbacks of multi-direct forcing algorithm for these situations are emphasized. A series of numerical Pluck tests were conducted to validate the fluid-dynamic forces calculation and structural responses of plates in multi-phase flows. The results were compared with experiments published in the literature, and good agreement was obtained.

Keywords LES · VOF-PLIC · FSI · Mindlin–Reissner plate · Adaptive mesh refinement

1 Introduction

Fluid–structure interaction (FSI) problems are characterized by one or more structures interacting with an internal and/or external fluid flow. Such problems play prominent roles in many scientific and engineering fields but a comprehensive study remains a challenge due to their strong non-linearity and multidisciplinary nature [1]. FSI is present in various forms and in different systems, being found from observations of nature [2, 3] to the search for perfection of man-made objects [4, 5].

Typical industrial applications of FSI are associated with complex structural geometries, moving boundaries, and high Reynolds number flows. Numerical approaches that simplify

the fluid or solid dynamics lack the physics needed to reproduce the field observations. This premise implies that both fluid and solid domains must be properly modeled and solved efficiently. As stated by Sotiropoulos [6], the most reliable solution would be to solve it as a monolithically coupled system. However, this approach requires sophisticated solvers to obtain the forces applied by the fluid and the corresponding vibration responses of the structure. The alternative is to solve the system by parts, having the fluid and solid solved with their numerical methods and coupled by boundary conditions.

The common computational cost drawback in FSI complex problems is in fluid calculations due to the complexity of the fluid dynamic mathematical model. The solver efficiency and robustness are found using Cartesian mesh instead of a traditional body-fitted unstructured grid. Besides the time-consuming mesh generation process, a moving boundary will imply the need for a re-meshing process that can be computationally expensive and require considerable coding effort. Cartesian mesh by itself can not be used to model and simulate a complex moving geometry.

Technical Editor Erick Franklin.

✉ Aldemir Ap Cavalini Jr
aacjunior@ufu.br

¹ Federal University of Uberlândia, Av. João Naves de Ávila, 2121, Sta. Mônica, Uberlândia, MG, Brazil

Over the past 4 decades, a simple and flexible method called immersed boundary (IBM), firstly proposed by Peskin [7], is being improved by countless authors to overcome this limitation.

Generally speaking on IBM, two different meshes are used: an Eulerian grid (Cartesian) to solve flow fields and a Lagrangian grid to force the solid boundary condition. There are several types of IBM. But, in this family of methods, velocities are interpolated from Eulerian to Lagrangian mesh; by knowing the velocity value at the immersed boundary, a force is calculated. This force is spread back to Eulerian mesh and used in Navier–Stokes (N-S) equation. The main differences between the methods arise from the calculation of the force step, which can be performed using implicit or explicit fluid solvers, and the interpolation/spread steps using diffuse or sharp techniques. Details about the different kinds of IBM can be found in several reviews about the subject [1, 6, 8–12].

However, complex flows with high Reynolds numbers are hardly simulated in feasible computational time using only Cartesian mesh and IBM. To overcome this challenge, the use of a block-structured mesh with adaptive refinement allows using efficient solvers due to Cartesian blocks and spending computational effort only in regions where it is needed due to the physical characteristics. Started by Roma et al. [13] with this goal, the now called MFSim code has been continually developed through the years into a multi-disciplinary code for multiphase flows [14, 15], reactive flows [16], and more recently FSI. These developments were done at the Fluid Mechanic Laboratory (MFLab), part of the School of Mechanical Engineering at Federal University of Uberlândia (UFU).

In the present paper, the LES filtering process is applied to the fluid dynamic model for incompressible, multi-phase, and isothermal flows of a Newtonian fluid. The turbulence closure is performed using the dynamic Germano [17] model. The multi-phase problem is modeled with VOF-PLIC method and the presence of a solid body was considered using multi-direct forcing IBM, developed by Wang et al. [18] based on Lima et al. [19]. This fluid model is also strongly coupled using the partitioned approach with the Mindlin–Reissner Plate (MRP) theory. This plate model was implemented in MFSim code in the context of the present work.

This paper aims to solve FSI problems involving multi-phase flows without added mass or artificial damping attached to the structural model, within feasible computational time. To evaluate fluid–structure interaction with the chosen methods, a series of numerical Pluck tests were performed and the natural frequencies of a vibrating plate were obtained. It was performed first in deep water, then in a water tank with the free surface close to the plate, matching Lindholm's experimental results [20]. It is worth mentioning that the novelty of

the present contribution relies on the solution of both fluid and structural models simultaneously, in which the forces applied to the structure by the fluid were obtained based on the resulting displacements and velocities of the structure, allowing to obtain, for instance, the free surface contour overtime of the fluid as a result of the structure movement.

It is worth mentioning that the hypothesis of incompressible flow was adopted, which implies that the mass of a fluid particle remains constant with time. However, gravitational effects can be taken into account using the Boussinesq approximation for non-isothermal flows. The incompressibility hypothesis is used for both phases in the modeling of two-phase flows, but the specific mass and viscosity vary in space according to the mapping made using a phase indicator function. In the MFSim platform, flows at high values of the Reynolds number are modeled using the filtered balance equations and sub-grid turbulence closure models for the LES methodology. Effectively, turbulent viscosity is about an order of magnitude greater than molecular viscosity. This gives the sensitivity of the simulations to the Reynolds number. The boundary layer transition to turbulence and roughness effects is still not modeled in the MFSim.

This paper is organized as follows: the mathematical modeling for both fluid and solid domains is presented in Sect. 2. Then, the numerical modeling for these equations is introduced in Sect. 3. After that, several validation test cases are discussed in Sect. 4 showing the generality of the chosen methods. Finally, closing remarks are presented in Sect. 5.

2 Differential algebraic model

2.1 Fluid dynamics

In a fluid system containing two immiscible and incompressible Newtonian fluids with an immersed body, as presented schematically in Fig. 1, the LES filtered continuity and linear momentum equations can be expressed for incompressible flows as:

$$\frac{\partial \bar{u}_i}{\partial x_i} = 0, \quad (1)$$

$$\rho \left[\frac{\partial \bar{u}_i}{\partial t} + \bar{u}_j \frac{\partial \bar{u}_i}{\partial x_j} \right] = - \frac{\partial \bar{p}}{\partial x_i} + \frac{\partial}{\partial x_j} \left[\mu \left(\frac{\partial \bar{u}_i}{\partial x_j} + \frac{\partial \bar{u}_j}{\partial x_i} \right) \right] - \frac{\partial \tau_{ij}^{SGS}}{\partial x_j} + f_i^{ext}, \quad (2)$$

where u_i is the velocity component, p is pressure, ρ is the fluid specific mass, μ is the dynamic viscosity, τ_{ij}^{SGS} is the sub-grid scale tensor and f_i^{ext} is the resulting external forces.

To capture the interface between two immiscible fluids, the VOF method from [21] is used and a third equation for

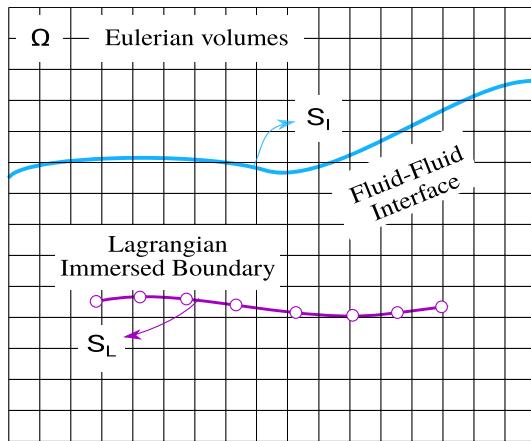


Fig. 1 Scheme of domain and boundaries

the volume fraction advection c is added to the system of equations:

$$\frac{\partial c}{\partial t} + \bar{u}_i \frac{\partial c}{\partial x_i} = 0. \quad (3)$$

where c represents a discrete color function field defining a fraction of the first phase in each volume $\mathcal{V}$, such that:

$$c = \frac{1}{\mathcal{V}} \int_{\Omega} \mathcal{H}(\mathbf{x}, t) d\mathbf{x} \quad (4)$$

where $\mathcal{H}$ is an indicator function defined as:

$$\mathcal{H}(\mathbf{x}, t) = \begin{cases} 1, & \text{if } \mathbf{x} \in \text{first phase} \\ 0, & \text{if } \mathbf{x} \in \text{second phase} \end{cases} \quad (5)$$

The volume fraction c is used to derive the physical properties ρ and μ in space and time using:

$$\varphi(\tilde{c}) \equiv \tilde{c} \varphi_1 + (1 - \tilde{c}) \varphi_2, \quad (6)$$

where φ is the property to be evaluated, subscripts denote the two phases, and $\tilde{c}$ is constructed by applying a smoothing spatial filter to c .

In this context, the external forces acting on a fluid particle are given by:

$$f_i^{ext} = f_i^g + f_i^\sigma + f_i^{ib}, \quad (7)$$

where $f_i^g = \rho g_i$, g_i is the gravity field, f_i^σ is the interface-tension force between phases on S_I , and f_i^{ib} is the force applied on the fluid particles by the solid body S_L . Both interface-tension and body force exist only in the interfaces, leading to:

$$f_i^\sigma(\mathbf{x}, t) = \int_{S_I} \sigma \kappa_I \mathbf{n}_I \delta(\mathbf{x} - \mathbf{x}_I) dS_I, \quad (8)$$

$$f_i^{ib}(\mathbf{x}, t) = \int_{S_L} F_i^{ib}(\mathbf{x}_L, t) \delta(\mathbf{x} - \mathbf{x}_L) dS_L, \quad (9)$$

where $\mathbf{n}_I$ is the normal vector on the interface, $\mathbf{x}$ is the fluid coordinate (Eulerian) vector, $\mathbf{x}_I$ is the phase interface coordinate (Lagrangian) vector, $\mathbf{x}_L$ is the solid body coordinate (Lagrangian) vector, F_i^{ib} is the body force in Lagrangian domain calculated with IBM method, σ is the interface-tension coefficient, and κ_I is the curvature evaluated using Eq. 10.

$$\kappa_I = -\frac{\partial \mathbf{n}_I}{\partial \mathbf{x}_I}. \quad (10)$$

In this case, the turbulence closure for τ_{ij}^{SGS} is modeled using the Dynamic Germano model [17].

The fluid–structure coupling is modeled using the Multi-Direct Forcing method, which consists of interpolating the fluid velocities neighboring the structure to calculate the force that the fluid exerts on it. This force is returned to the fluid, which is taken into account in the solution of the fluid dynamics. A strong coupling process is used. This process consists of making an interactive procedure between fluid and structure until convergence is established. Thus, it is guaranteed that the displacements of the structure and fluid particles that are cohesive to it are equal. As for the fluid–fluid interface modeling, the VOF method is used to reconstruct the interface at each time step.

2.2 Solid dynamics

The movement of a moderated thick plate can be modeled using a simplification of the Mindlin–Reissner plate theory, as proposed by Auricchio [22], with infinitesimal kinematics, expressed as:

$$\xi(x, y, z) = z \theta_y(x, y), \quad (11)$$

$$\eta(x, y, z) = -z \theta_x(x, y), \quad (12)$$

$$\omega(x, y, z) = \omega(x, y), \quad (13)$$

where ξ , η , and ω are the plate's displacement along x , y , and z axis, respectively; θ_x and θ_y are the rotations of the transverse line elements (initially perpendicular to the mid-surface).

As the plate mid-surface in its undeformed configuration remains straight, but not necessarily normal to the deformed mid-surface, the curvature κ and shear strain Γ can be expressed as:

$$\kappa = \mathbf{L} \theta \text{ and } \Gamma = [\mathbf{e} \theta + \nabla \omega], \quad (14)$$

where $\mathbf{L}$, ∇ , and $\mathbf{e}$ are defined as:

$$\mathbf{L} = \begin{bmatrix} 0 & \partial/\partial x \\ -\partial/\partial y & 0 \\ -\partial/\partial x & \partial/\partial y \end{bmatrix} \quad (15)$$

$$\mathbf{e} = \begin{bmatrix} 0 & 1 \\ -1 & 0 \end{bmatrix}, \quad \nabla = \left\{ \frac{\partial}{\partial x}, \frac{\partial}{\partial y} \right\}, \text{ and } \theta = \begin{Bmatrix} \theta_x \\ \theta_y \end{Bmatrix}.$$

The plate's material is assumed to be homogeneous, isotropic, and linearly elastic. Thus, the dynamic behavior of the plate can be represented by the following differential equation:

$$\mathbf{M} \ddot{\mathbf{X}} + \mathbf{K} \mathbf{X} = \mathbf{F}^{ib}, \quad (16)$$

where $\mathbf{X}$ is the displacement vector $\{\xi, \eta, \omega\}^T$, $\ddot{\mathbf{X}}$ is the second derivative in time, $\mathbf{M}$ and $\mathbf{K}$ are the mass and stiffness matrices, respectively, obtained using the simplifications above and the chosen finite element method, as detailed in the next section.

3 Numerical and computational modeling

The FSI problem will be solved in the present work using a partitioned approach with strong coupling. By solving the associated problem this way, the computational resources can be used more efficiently in terms of parallelism, mainly due to the pronounced difference in discretization and computational time spent to solve the fluid and solid equations.

It is important to note that by solving each domain separately with appropriated numerical methods, the authors find the programming to be cleaner, less prone to bugs, and keep the code problem-independent. Besides, debug and validation can be performed separately in the early stages restricting the further FSI problems to the coupling algorithm.

3.1 Fluid model

To solve Eqs. 1 and 2 for the fluid domain, a projection method was used and the time discretization was performed with Implicit-Explicit (IMEX) methods. In space, the finite volume approach was used and the discretization with Barton's method for monotonic transport as presented in [23], applied on an in-house code with Structured Adaptive Mesh Refinement (SAMR) based on [24]. It is important to state that the mesh changes during the simulation refining and coarsening regions according to a given criterion.

Adaptive mesh refinement has been adopted in the present work looking for the computational efficiency in fluid–structure interaction problems in two-phase flows. Furthermore, parallel processing with dynamic load balancing has been one of the focuses in the research for greater computational efficiency.

Figure 2 shows the mesh used to a given high Reynolds simulation using MFSim. The refinement is established using two different criteria, plate's position, and fluid's vorticity magnitude.

The imposition of non-slip condition at the boundary of the arbitrarily shaped body is done by multi-direct forcing IBM [18] using a compact support delta function $\mathcal{D}$ replacing the Dirac delta function δ from Eq. 9 as:

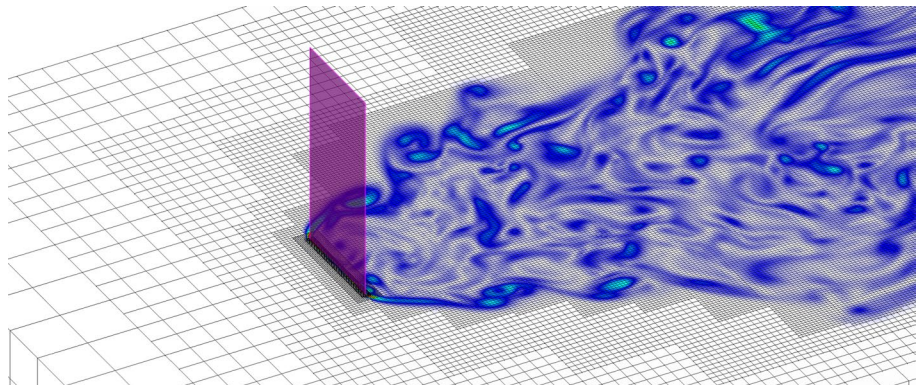
$$\mathcal{D}(\mathbf{x} - \mathbf{x}_L) = \frac{1}{h^3} \phi\left(\frac{x - x_L}{h}\right) \phi\left(\frac{y - y_L}{h}\right) \phi\left(\frac{z - z_L}{h}\right), \quad (17)$$

where h is the mesh spacing and the function ϕ is used based on [13] that enumerate several advantages of this choice:

$$\phi(r) = \begin{cases} \frac{1}{6}(5 - 3|r| - \sqrt{-3(1 - |r|)^2 + 1}), & 0.5 \leq |r| \leq 1.5 \\ \frac{1}{3}(1 + \sqrt{-3r^2 + 1}), & |r| \leq 0.5 \\ 0 & \text{otherwise.} \end{cases} \quad (18)$$

The volume fraction advection Eq. 3 is solved using VOF-PLIC method based on [25] and performed in two steps:

Fig. 2 Example of a seven levels mesh adaptivity on a flow through a transversal plate, simulated with MFSim



an interface reconstruction using PLIC method followed by its advection with a Lagrangian algorithm. Additionally, to obtain the curvature and surface tension force, the normal is calculated as a normalized gradient of volume fraction field c using the discretization proposed by Shirani et al. [26].

3.2 Solid model

Solving Eq. 16 needs the mass matrix, stiffness matrix, and body force vector. In the present work, those variables are calculated using the finite element method with the quadrilateral four-node element, based on linked interpolation and the mixed approach (Q4-LIM) [22]. In their study, Auricchio and Taylor [22] developed shape functions for the following functional:

$$\Pi(\omega, \theta, S) = \frac{1}{2} \int_A [\mathbf{K}^T(\theta) \mathbf{D}_B \mathbf{K}^T(\theta)] dA - \frac{1}{2} \int_A [\mathbf{S}^T \mathbf{D}_S^{-1} \mathbf{S}] dA + \int_A [\mathbf{S}^T (\nabla \omega + \mathbf{e}\theta)] dA + \Pi_{ext}, \quad (19)$$

where A is the area of the element, Π_{ext} represents the loads and boundary effects, $\mathbf{D}_B$ and $\mathbf{D}_S$ come from the constitutive relations for homogeneous, linearly elastic and isotropic material given by Eqs. (20) and (21). Therefore, nonlinear effects due to, for instance, large deformations, are disregarded in this model. Details about this assumption can be observed in Auricchio and Taylor [22].

$$\mathbf{D}_B = D \begin{bmatrix} 1 & \nu & 0 \\ \nu & 1 & 0 \\ 0 & 0 & \frac{1}{2}(1 - \nu) \end{bmatrix} \quad \text{where } D = \frac{E h^3}{12(1 - \nu^2)}, \quad (20)$$

$$\mathbf{D}_S = k G h \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} \quad \text{where } G = \frac{E}{2(1 + \nu)}. \quad (21)$$

in which E is the Young's modulus, ν is the Poisson's ratio, and k is the correct factor for non-constant shear stress set as $\pi^2/12$.

The fields ω , θ , and S are approximated by:

$$\omega = N_\omega \hat{\omega} + N_{\omega\theta} \hat{\theta}, \quad (22)$$

$$\theta = N_\theta \hat{\theta} + N_b \hat{\theta}_b, \quad (23)$$

$$S = N_S \hat{S}, \quad (24)$$

where $\hat{\omega}$, $\hat{\theta}$, $\hat{\theta}_b$, and $\hat{S}$ are the degrees of freedom for translation, rotation, internal rotational, and shear stress, respectively; N_ω , $N_{\omega\theta}$, N_θ , N_b , and N_S are the set of shape functions.

Upon introducing the approximations into Π , it is possible to reduce the resultant algebraic system by eliminating the internal rotation and the shear stress. Therefore, a stiffness matrix that is dependent only of $\hat{\omega}$ and $\hat{\theta}$ is obtained. For further details of this process, see the work of Auricchio and Taylor [22]. The mass matrix and load vector are consistent with the proposed shape functions without the internal rotation and the shear stress parameters.

With no non-linearities on the structural model and the assumption that mode shapes are not affected by the surrounding fluid, a modal transformation is done by solving an eigenvalue problem and converting the finite element matrices to the modal domain. At each fluid time step, an integration using Runge–Kutta–Dormand–Prince [27] is performed. After integration, the resulting plate velocities and displacements are transformed back to physical space.

3.3 Coupling

The coupling algorithm between the fluid and solid domains is performed by iterating the Lagrangian force calculation from IBM and the finite element calculations of displacements and positions of the boundary. It is based on the fluid–structure coupling procedure of [28].

The advantages of the present chosen methods are: first, Lagrangian force calculations depend on the position and velocity of the solid. An interpolation of the predicted velocity from the projection method is also used. Second, the finite element displacements and positions depend only on the Lagrangian forces.

To apply the strong coupling, a loop between those two calculations is needed. As the reader might have noted by now, no equation needs to be solved in the fluid domain during the coupling. Therefore, the algorithm is faster compared to those where a Poisson equation is solved at each iteration [29, 30].

The adopted fluid–structure coupling procedure in the one-time step of the MFSim code is presented in Algorithm 1.

Algorithm 1 MFSim time step

```

Calculate the time step size
Perform VOF-PLIC operations:
    Solve the advection equation
    Calculate the curvature
    Calculate the interface-tension force
Update the physical properties ( $\rho$ ,  $\mu$ )
Solve the velocity prediction without body force ( $\mathbf{u}^*$ )
while residue < tolerance do
    Perform some iterations on IBM operations:
        Interpolate the Eulerian velocities to IB points
        Calculate the Lagrangian Force ( $F^{ib}$ )
        Spread the force to Eulerian domain ( $f^{ib}$ )
    Spread the Lagrangian force to finite element model
    Calculate the structural displacement and velocities
    Move the IB points and update its velocities
     $residue = F^{ib} - F_{old}^{ib}$ 
end while
Correct the velocity prediction with Eulerian force ( $\mathbf{u}^{**}$ )
Solve the Poisson equation for pressure correction
Correct the pressure and velocity

```

3.4 Adaptive mesh refinement

With the AMR algorithm, a locally refined mesh is achieved in regions where the error in the solution is above some threshold previously specified. This is especially useful for fluid–structure simulations to maintain the accuracy of force calculations without penalizing the whole solver with an overly refined mesh.

AMR strategies may be employed in Cartesian grids via octree or structured adaptive mesh refinement [24]. In the used SAMR approach, cells located at a given refinement level are clustered into Cartesian blocks called patches. These patches are organized in a hierarchy of Cartesian grids with different levels of refinement which cover the entire domain.

A level of refinement comprises one or more patches that do not intersect each other and share the same grid spacing. These levels are always organized in a nested configuration to avoid using information from an overly coarse cell in the fine grid.

The fluid discretizations (in time as well as in space) present a second-order accuracy. Therefore, the interpolations between levels are done using second-order Lagrange polynomials to guarantee the whole calculation in this accuracy.

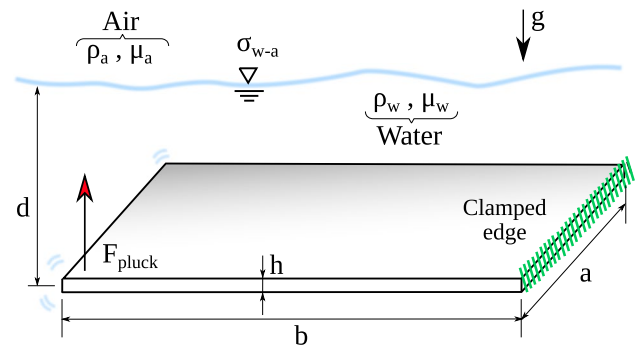
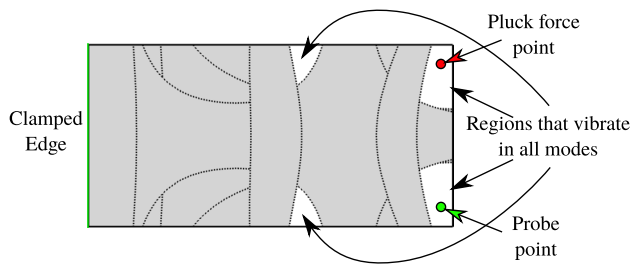
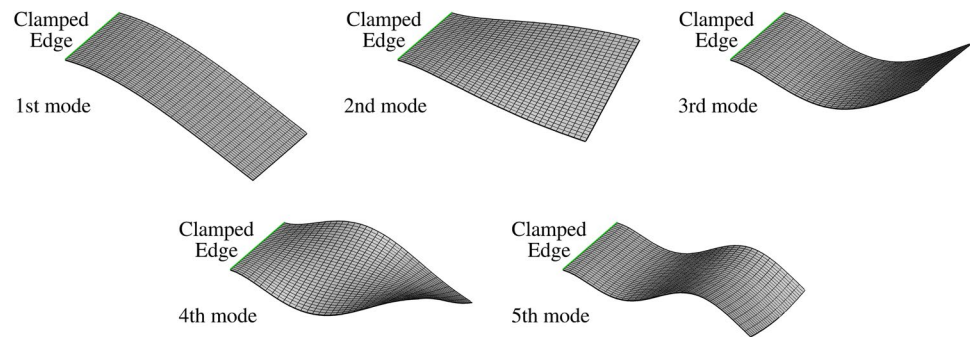


Fig. 3 Scheme of studied problems

4 Fluid–structure interaction

To demonstrate the potentiality of the MFSim code to simulate complex flows with fluid–structure interaction, two problems were proposed in the present work based on Lindholm’s consolidated experiment [20] on submerged plates in water.

In the first set of tests, the distance d shown in Fig. 3 will be very large and the water surface influence on results can be neglected. A different set of plate dimensions (a , b , and h) were tested in this case. For the second

Fig. 4 First five vibrating modes for a plate with $b = 2a$ **Fig. 5** Force application and displacement probe locations

set of simulations, the gap d between the plate and the water/air interface is modified and the plate dimensions were fixed.

For all simulations, the fluid is initially at rest and the plate is deformed by its static own weight. The analyses were performed by measuring the natural frequencies from the plate's FRF responses.

In both problems, the surrounding fluid is viscous, homogeneous, Newtonian, and incompressible. The fluids used were air ($\rho_a = 1.225 \text{ kg/m}^3$ and $\mu_a = 1.8375 \times 10^{-5} \text{ Pa s}$) and water ($\rho_w = 998.2 \text{ kg/m}^3$ and $\mu_w = 1.002 \times 10^{-3} \text{ Pa s}$). The interface tension is $\sigma_{w-a} = 0.07286 \text{ N/m}$, and gravity $g = 9.81 \text{ m/s}^2$. Only steel plates were tested in the present work ($E = 2.068 \times 10^{11} \text{ Pa}$, $\rho_s = 7830 \text{ kg/m}^3$, and $\nu = 0.3$).

4.1 Numerical hammering

In both problems, an impact force discrete in time was applied at a given point of the cantilever plate. Additionally, the displacement of the plate was measured with a probe at each time step.

A procedure was established to find the best places to measure the displacement and apply the force in each simulated case. First, the mode modes of the tested plate were calculated. The first five modes shapes for a plate with $b = 2a$ are shown in Fig. 4.

Second, the mode shapes were plotted in 2D using the displacement w as a contour. Then, a blanking operation is applied to remove the parts with large displacements.

The third step is an overlap of the described plot for the selected modes in a single figure. The result is a map indicating the regions where the plate will vibrate in all selected modes (blank spaces). Figure 5 is an example of this map. In this case, four regions have significant vibration amplitude on all selected mode shapes. The measurement and force locations are also indicated in Fig. 5. It is worth mentioning that the force must be applied and the probe placed in the mentioned regions to observe evident vibration resonances (peaks in Fig. 8) at the natural frequencies associated with the vibration modes presented in Fig. 4.

It is important to notice that the choice of a discrete force application can be troublesome with loose FSI coupling due to stability issues. But the advantage of being discrete is that this force will always excite all frequencies, and no additional attention is needed in FRF signals.

4.2 Deep water analysis

In this first set of cases, the experiment [20] kept the distance d to the free surface as far as possible to avoid any influence on the results. Therefore, in these simulations, the domain is completely filled with water and the VOF-PLIC model is not being used.

The size of the computational domain was maintained for all the test cases and has the dimension parametrized by the plate width a , as given by Fig. 6(a). The mesh used in these cases has the finest delta of $6.36 \times 10^{-3} \text{ m}$ and 5 levels of refinement as can be seen in Fig. 6(b). The boundary conditions were a wall type (Dirichlet zero velocities) except for the upper boundary where a Neumann condition is applied for velocities and the gauge pressure is set zero.

The time step is variable to maintain a CFL of 0.5, but it stabilizes along with the simulation around 10^{-4} s . During several tests performed, it was observed that large time steps could be used, but that also would diminish the frequency of fluid force calculations and, therefore, filter some excitations to the structural model. In extreme cases, this could lead to severe smoothing on the natural frequency peaks calculated.

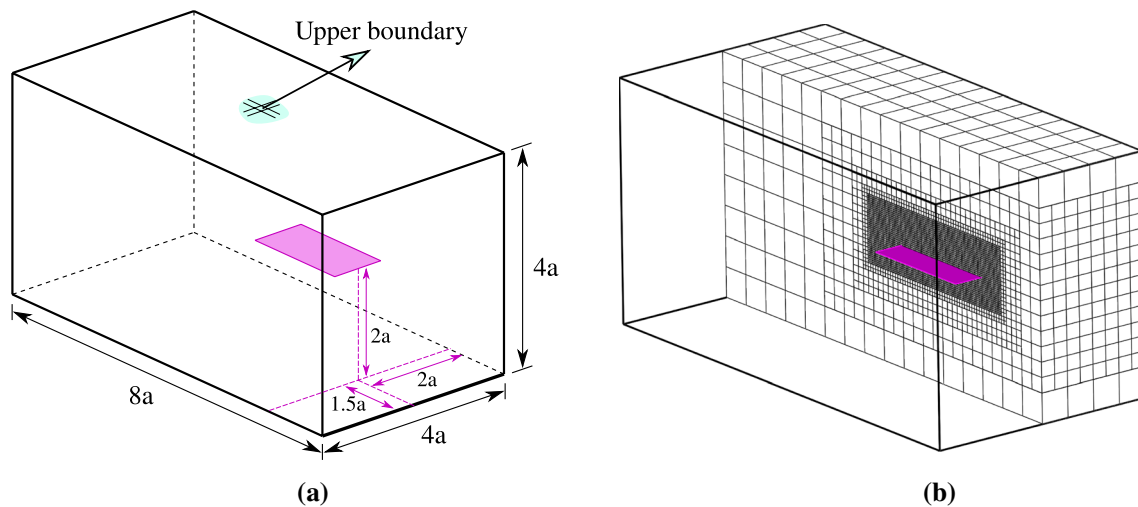


Fig. 6 (a) Computational domain and (b) mesh configuration

Table 1 Plate dimensions used in the deep water simulations setup [20]

Case	a [mm]	b [mm]	h [mm]
1.1	203.2	203.2	2.66192
1.2	203.2	406.4	2.66192
1.3	203.2	609.6	2.66192
1.4	203.2	203.2	4.83616
1.5	203.2	406.4	4.83616
1.6	203.2	609.6	4.83616
1.7	203.2	1016.0	4.83616

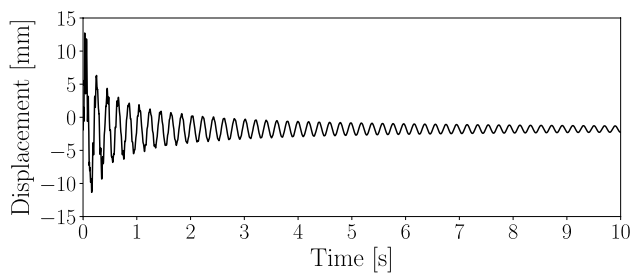


Fig. 7 Plate displacement through thickness direction measured in case 1.2

A total of seven cases were simulated changing the plate dimensions as in the experiments of Lindholm [20]. Table 1 shows the test cases adopted in the present contribution, as well as by Lindholm [20].

The cases were simulated until 10 seconds, along with the displacement signals were acquired for a probe point in the plate (see Fig. 5) at each time step. An example of this signal is given in Fig. 7 for case 1.2. The Mindlin–Reissner theory is devoted to representing the linear behavior of thick

plates. Thus, its main limitation is associated with nonlinear responses due to, for instance, large deformations. However, only linear vibration responses are expected regarding the analyses performed in the present contribution (see Fig. 7).

A Frequency Response Function (FRF) was calculated for every test case using the Discrete Fourier Transform (DFT) from the impact force and displacement signals (see the force location in Fig. 5).

Due to the variable time step, a re-sampling was needed before the DFT can be applied. The new sampling was done by linear interpolation of the signal with the smallest time step found in the case. After this, the frequencies present can be found and compared with the experimental data. A collection of FRFs are shown in Fig. 8 for all seven cases and the peaks are labeled for easier identification.

These frequencies are presented in Tables 2, 3, 4 and 5, in their dimensionless form, and compared with experimental results conducted by Lindholm et al. [20], as well as other numerical studies of Liang et al. [31], Fu and Price [32], Ergin and Ugurlu [33] and Marcus [34].

The errors between numerical and experimental measurements are also indicated. The experimental tests were performed in a 1.83x3.66x2.44 m deep water tank with the plate being clamped to rigid beam support attached to the tank. The level of the free water surface with respect to the plate was controlled by the depth of water in the tank. The actuator and sensor devices employed in the experiments were chosen due to their non-contact operation avoiding adding additional mass to the vibrating plate. Excitation was obtained by using an electromagnet located directly beneath the plate. A noncontact displacement sensor was used to measure the vibration responses of the plate. Equation 25 demonstrates how the obtained natural frequencies were converted to its dimensionless form ω_o .

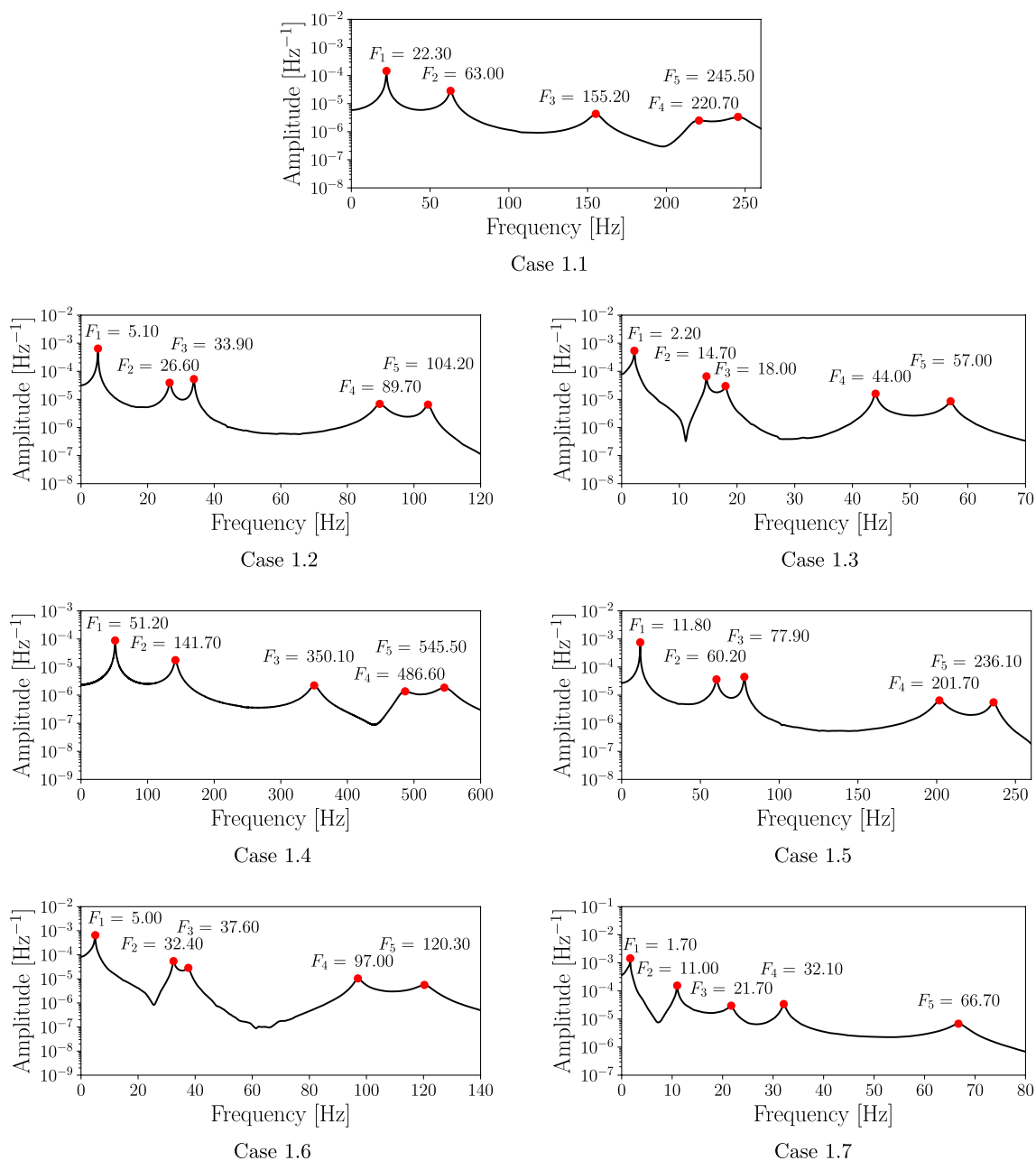


Fig. 8 Natural frequencies in deep water

$$\omega_o = 2\pi F a^2 \sqrt{\frac{\rho_s h}{D}} \quad (25)$$

where F is the natural frequency obtained from Fig. 8.

Note that the errors between experiments and simulations demonstrated to be small. The maximum error found in the present work was 12.51% in the 5th frequency of case 1.7, Liang et. al [31] has a 15.74% error in the 4th frequency of case 1.3, Fu and Price [32] and Ergin and Uğurlu [33] present 10.66% error in 1st frequency of case 1.4 and Marcus [34] show 15.84% error in the 5th frequency of case 1.2.

This can be associated with some differences between the model and the real system that were not mentioned in the reference works.

4.3 Shallow water analysis

In this case, no modifications were made in the domain, boundary conditions, or mesh resolution used to simulate the cases where plate distance from water/air interface surface changes. But now, since the distance from the free surface is small, it

Table 2 Deep water tests (cases 1.1 and 1.2)

Frequencies	Case 1.1			Case 1.2			
	Lindholm	MFSim	Liang	Lindholm	MFSim	Liang	Marcus
1st	1.478	1.398	1.415	1.305	1.278	1.296	1.278
		5.43%	4.27%		2.00%	0.69%	2.03%
2nd	4.345	3.948	4.319	7.602	6.668	7.571	7.117
		9.14%	0.60%		12.29%	0.42%	6.38%
3rd	10.014	9.726	8.621	8.772	8.498	8.057	8.145
		2.87%	13.91%		3.12%	8.15%	7.15%
4th	14.839	13.831	14.942	25.387	22.485	22.604	22.504
		6.79%	0.70%		11.43%	10.96%	11.36%
5th	16.889	15.385	18.056	26.828	26.120	26.338	22.579
		8.90%	6.91%		2.64%	1.83%	15.84%

Table 3 Deep water tests (cases 1.3 and 1.4)

Frequencies	Case 1.3			Case 1.4				
	Lindholm	MFSim	Liang	Lindholm	MFSim	Liang	Ergin	Fu
1st	1.284	1.241	1.246	1.795	1.766	1.791	1.986	1.986
		3.37%	2.96%		1.60%	0.23%	10.66%	10.66%
2nd	8.585	8.291	7.792	5.288	4.888	5.548	5.568	5.466
		3.43%	9.23%		7.57%	4.93%	5.30%	3.37%
3rd	11.493	10.152	11.238	12.420	12.076	10.914	13.580	13.631
		11.66%	2.22%		2.77%	12.13%	9.34%	9.75%
4th	25.992	24.817	21.900	18.517	16.785	19.811	18.792	19.029
		4.52%	15.74%		9.35%	6.99%	1.48%	2.77%
5th	36.732	32.149	35.348	20.408	18.816	21.515	21.355	21.304
		12.48%	3.77%		7.80%	5.42%	4.64%	4.39%

Table 4 Deep water tests (cases 1.5 and 1.6)

Frequencies	Case 1.5			Case 1.6		
	Lindholm	MFSim	Liang	Lindholm	MFSim	Liang
1st	1.678	1.628	1.654	1.605	1.552	1.598
		2.96%	1.42%		3.29%	0.42%
2nd	9.348	8.306	9.741	10.510	10.058	9.978
		11.15%	4.20%		4.30%	5.07%
3rd	11.155	10.748	10.289	13.084	11.673	13.628
		3.64%	7.76%		10.79%	4.15%
4th	30.966	27.830	28.870	31.089	30.113	28.047
		10.13%	2.63%		3.14%	9.79%
5th	33.758	32.576	31.781	41.224	37.347	42.860
		3.50%	14.48%		9.41%	3.97%

will affect the solution and, therefore, the VOF-PLIC model is activated to capture and treat the water/air interface dynamics.

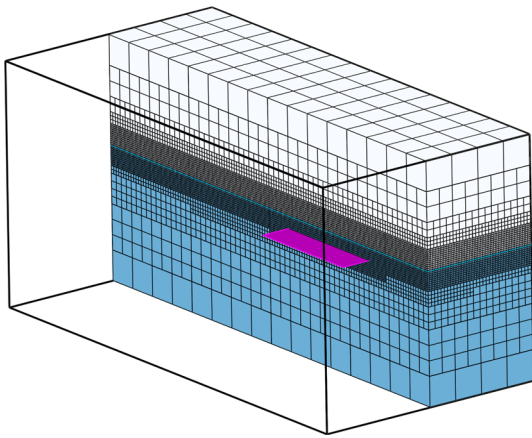
This interface needs to be well resolved to obtain good results in the simulations. To do that, the mesh refinement criteria will have to include this variable along with the plate's position. The mesh configuration for these two-phase flow cases can be seen in Fig. 9 where the blue contour represents the water part.

A total of five two-phase flow cases were simulated with different distances from water surface d as can be seen in Table 6. In all these cases, the plate geometry from case 1.2 was used.

Initially, in these two-phase flow simulations, the fluid is at rest with the pressure field equal to the hydrostatic pressure distribution, and the plate is deformed by its static own weight. Following the previous methodology, a numerical

Table 5 Deep water tests (case 1.7)

Frequency	Lindholm	MFSim	Liang	Marcus	Ergin
1st	1.552	1.466 5.57%	1.552 0.04%	1.466 5.59%	1.568 1.02%
2nd	10.014	9.486 5.27%	9.699 3.15%	9.397 6.16%	10.064 0.51%
3rd	20.943	18.713 10.65%	21.475 2.54%	19.311 7.79%	20.680 1.25%
4th	29.237	27.681 5.32%	27.217 6.91%	27.674 5.35%	29.564 1.12%
5th	65.747	57.519 12.51%	65.735 0.02%	59.399 9.65%	63.686 3.13%

**Fig. 9** Mesh configuration for the two phase flow cases**Table 6** Shallow water simulations setup

Case	Plate depth [mm]
2.1	203.20
2.2	121.92
2.3	81.28
2.4	60.96
2.5	40.64

hammering discrete in time is applied to the plate and the flow field and plate's position are monitored.

To obtain good agreement with the experiment when the plate is too close to the free surface, the same cases taken in the material experiment must be taken for the numerical tests. Lindholm's [20] work kept the maximum plate displacement around 10 mm to avoid the surface waves greatly disturbing the plate's measurements.

The first test in case 2.5 was made to investigate these limitations. The hammering force applied in this test generated almost 60 mm of displacement on the plate, therefore, passing the distance from free surface $d = 40.64$ mm.

A highly disturbing free surface and its effect on the plate displacement signals were observed.

Figure 10 shows a slice taken from the field close to the plate's free edge and the time evolution of the free surface in this test case can be seen. It is presented the visualization of fluid dynamics promoted by the movement of the plate. Initially, it was positioned entirely submerged, at a depth of $d = 40.64$ mm. Plate movement promotes liquid movement and even entrainment of air bubbles. It is noticed that these air bubbles touch the ends of the plate, which shows that the method allows the interaction of the plate to occur simultaneously with the two phases, that is, liquid and gas.

In this case, a mesh independence study was not carried out. However, the use of adaptive refinement enabled a good agreement of the computational results with the results of material experiments, concluding that the refinement used is sufficient. Five levels of refinement were enough to a good representation of the liquid–gas interface, allowing even the capture of air bubble entrainment, as can be observed in Fig. 10.

Using a smaller hammering force, to keep the plate's displacement below the 10 mm limitation, barely any movement can be seen in the free surface. The difference between the DFT signals from plate displacement when its maximum value is 60 mm and 10 mm are presented in Fig. 11. The natural frequencies can be seen in both signals, but the peaks are better pronounced when the interface is not disturbed much. Therefore, all two-phase flow simulations presented here were made avoiding to pass this limit.

The same process of re-sampling from previous cases was performed here. But, the frequency results are presented in an increasing percentage from deep water frequency and plotted against the water distance d nondimensionalized by plate width a . A figure is presented for each vibrating mode comparing the results with the experiment and the numerical simulations of Marcus [34]. Following the mode numbering done in Lindholm's work [20], the 4th vibrating mode is not present for this plate configuration. Therefore, the five first frequencies consist of modes 1, 2, 3, 5, and 6 (see Fig. 12).

The present results agree with experiments, and in several points better than Marcus [34]. This example shows that the simple fact of changing the fluid-specific mass around the plate and capturing the water/air interface is responsible for obtaining the expected behavior.

5 Conclusion

The incompressible fluid–structure interaction problem with multi-phase flows was solved in the present contribution using the partitioned approach and adaptive mesh refinement. Corrections, such as added mass or artificial damping,

Fig. 10 Free surface contour over time (case 2.5 of Table 6; plate depth of 40.64 mm)

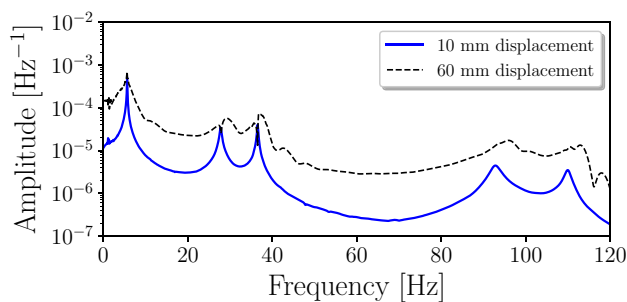
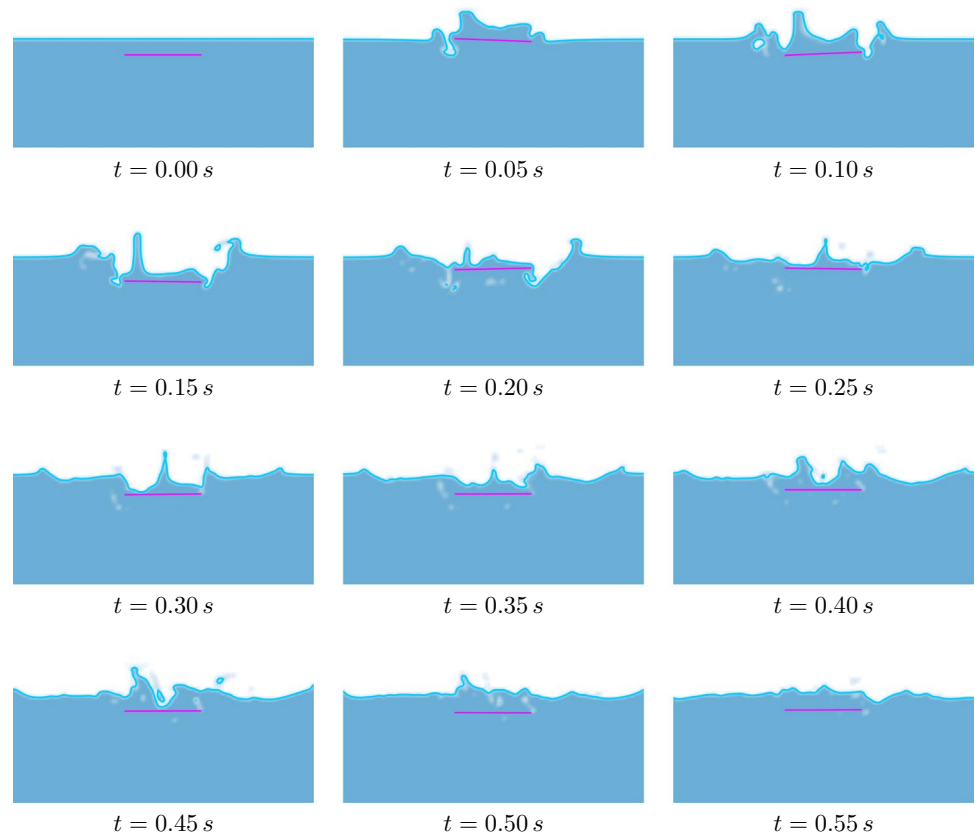


Fig. 11 Natural frequencies in case 2.5 tests

attached to the structural model were not adopted within feasible computational time. Various numerical Pluck tests were performed to evaluate fluid–structure interaction with the chosen methods. The natural frequencies of a vibrating plate were obtained. The numerical results were compared

with experimental tests performed first in deep water, then in a water tank with the free surface close to the plate. Satisfactory results were obtained in all evaluated test cases. As mentioned, the novelty of the present contribution relies on the solution of both fluid and structural models without any empirical corrections and simplifying assumptions, allowing to obtain, for instance, the free surface contour overtime of the fluid as a result of the structure movement. Therefore, the present method of FSI simulation can be applied without corrections in the structural model (such as added mass or artificial damping). Therefore, a simple change of boundary and initial conditions in the fluid are enough to capture some troublesome problems of structural modeling, like the proximity to the water tank floor and other vibrating plates, or a turbulent flow around the plate. Future work will be dedicated to evaluating the considered FSI simulation in more complex submerged structures, facing the obtained results with experimental data.

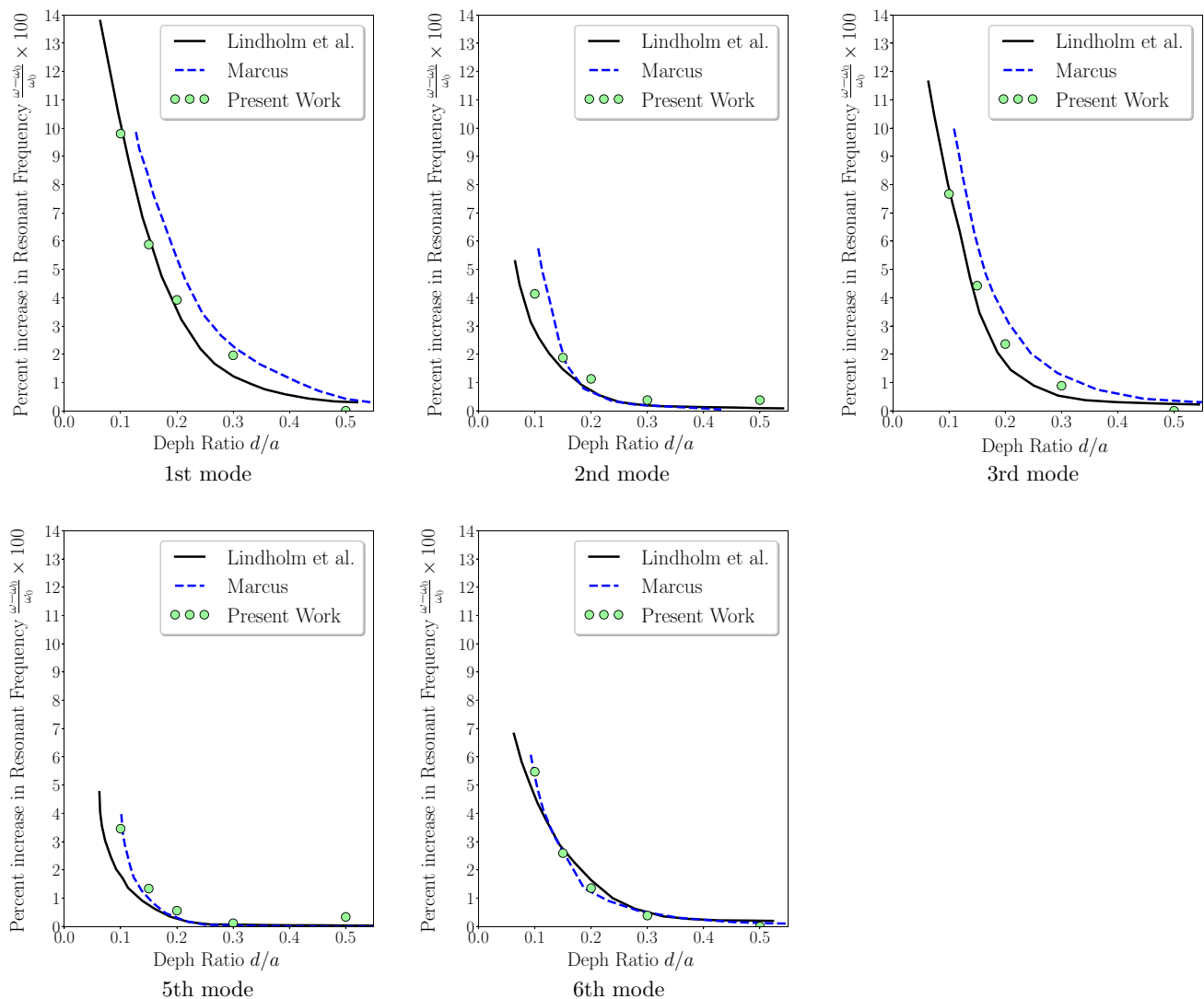


Fig. 12 Natural Frequencies increase with plate depth

Acknowledgements The authors are thankful for the financial support provided to the present research effort by CNPq (306200/2017-1, 431337/2018-7, 305958/2020-8, and 431337/2018-7), FAPEMIG (PPM-00187-18), CAPES through the INCITEIE, and PETROBRAS (2018/00082-0, 2018/00419-5, 2019/00056-2, and 2019/00106-0).

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